

## Seunghan Lee

- Date of Birth: July 29, 1996
- Country: Republic of Korea
- Military Service: Completed
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  - Blog: seunghan96.github.io (1400+ AI-related posts)

## Careers (Brief)

**LG AI Research**: December 2025 –

- **Role**: AI Scientist
- **Division**: Data Intelligence Lab > Business Intelligence AI (BIAI)

**KRAFTON**: May 2025 – December 2025

- **Role**: Deep Learning Engineer/Researcher
- **Division**: Deep Learning Division > AI Research > AI Agent Dept. > User Modeling AI Team

**Naver Cloud (internship)**: March 2025 – May 2025

- **Role**: AI Engineer/Researcher (Internship)
- **Division**: HyperClova X

**SK Telecom (internship)**: July 2019 – August 2019

- **Role**: Data Scientist (Internship)
- **Division**: ICT Infra Center > Data Science > Data Analytics (AI Home)

## 1. Education

**Undergraduate**

**Yonsei University, Seoul** (March 2015 – August 2020)

- **Majors**: Business Administration / Applied Statistics
- **GPA (4.5 scale)**: Overall 4.19 (Business 4.31, Applied Statistics 4.35)
- **Additional Information**:
  - Early graduation (7 semesters)
  - Summa Cum Laude (Top 1% of the cohort)

**Integrated Master's & PhD Program**

**Yonsei University, Seoul** (September 2020 – August 2025)

- **Major**: Statistics and Data Science
- **GPA (4.5 scale)**: 4.29
- **Co-advisors**:
  - Prof. Taeyoung Park (DSLab)
  - Prof. Kibok Lee (ML Lab)

## 2. Research Interests

- **Deep Learning for Time Series**
  - Multimodal Time Series Forecasting
  - Agentic Time Series Forecasting
  - Time Series Foundation Models
  - Representation Learning
  - Others (e.g., Self-Supervised Learning, Diffusion Models, Anomaly Detection)
- **AI for Finance**
  - Financial Time Series
  - Financial Language Models
- **Deep Learning for Tabular / Graphs**

## 3. Publications

([*]Equally Contributed, [†]Co-corresponding Authors)

### 29. Implied Volatility Surface Forecasting Under Earnings-Driven Distribution Shift: Transfer from Ordinary Trading Days (2026)

- Hyunoh Yeo, Jaehoon Lee, **Seunghan Lee**, Jun Seo, Junhyeok Kang, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Sangjun Han, Dongwan Kang, Hwanil Choi, Soonyoung Lee, Wonbin Ahn
- In *ICAIFW (AI-Driven Market Microstructure in Centralized and Decentralized Finance), 2026. [TBD]*

### 28. OneShot: Label-Efficient Detector Weighting for Time Series Anomaly Detection (2026)

- **Seunghan Lee**, Jun Seo, Jaehoon Lee, Junhyeok Kang, Sangjun Han, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Soonyoung Lee, Wonbin Ahn
- In *arXiv preprint, 2026. [arxiv:2610.16219]*

### 27. Quantile Alignment Improves Zero-Shot Time-Series Foundation Models (2026)

- Junhyeok Kang, Sangjun Han, Jun Seo, **Seunghan Lee**, Jaehoon Lee, Dongwan Kang, Minjae Kim, Sungdong Yoo, Tae Yoon Lim, Hwanil Choi, Soonyoung Lee, Wonbin Ahn
- In *NeurIPSW (FMTS), 2026. [TBD]*

### 26. Composite Kernel Decomposition of Gaussian Process for Time-Series Foundation Models (2026)

- Sangjun Han, Hyunoh Yeo, Joanie Hayoun Chung, Junhyeok Kang, **Seunghan Lee**, Jaehoon Lee, Jun Seo, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Minjae Kim, Sungdong Yoo, Soonyoung Lee, Wonbin Ahn
- In *NeurIPSW (FMTS), 2026. [TBD]*

### 25. EXAONE Demand 1.0: A Time Series Foundation Model for Demand Forecasting (2026)

- **Seunghan Lee**, Sangjun Han, Jun Seo, Junhyeok Kang, Jaehoon Lee, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Minjae Kim, Sungdong Yoo, Soonyoung Lee, Wonbin Ahn
- In *Technical report, 2026. [arxiv:2609.30880]*

### 24. PALM: Point-in-Time Adaptation for Financial Language Models (2026)

- **Seunghan Lee**, Jun Seo, Jaehoon Lee, Junhyeok Kang, Sangjun Han, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Soonyoung Lee, Wonbin Ahn
- In *ICAIFW (LLMs and Generative AI for Finance), 2026. [arxiv:2609.30316]*

### 23. EXAONE Finance 1.0: An Attention-free Time Series Foundation Model for Financial Time Series (2026)

- **Seunghan Lee**, Jaehoon Lee, Jun Seo, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Minjae Kim, Sungdong Yoo, Junhyeok Kang, Sangjun Han, Soonyoung Lee, Wonbin Ahn
- In *Technical report, 2026. [arxiv:2609.04239]*

### 22. Explaining Time Series Forecasting with Horizon-Resolved Attribution (2026)

- **Seunghan Lee**, Jun Seo, Jaehoon Lee, Junhyeok Kang, Sangjun Han, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Soonyoung Lee, Wonbin Ahn
- In *NeurIPSW (XAI4Science), 2026. [arxiv:2609.12639]*

### 21. FinVerse: Financial Time-Series Benchmark (2026)

- Jaehoon Lee, Jun Seo, **Seunghan Lee**, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Minjae Kim, Sungdong Yoo, Junhyeok Kang, Sangjun Han, Soonyoung Lee, Wonbin Ahn
- In *Technical report, 2026. [arxiv:2608.03259]*

### 20. ReasonCast: Towards Explainable Time Series Forecasting with Reasoning (2026)

- **Seunghan Lee**, Jun Seo, Jaehoon Lee, Junhyeok Kang, Sangjun Han, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Soonyoung Lee, Wonbin Ahn
- In *arXiv preprint, 2026. [arxiv:2608.01875]*

### 19. Beyond Magnitude and Shape: A Direction-Aware Loss for Time Series Forecasting (2026)

- **Seunghan Lee**, Jaehoon Lee, Jun Seo, Junhyeok Kang, Sangjun Han, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, Soonyoung Lee, Wonbin Ahn
- In *arXiv preprint, 2026.* [arxiv:2608.01857]

### 18. When Summaries Distort Decisions: Information Fidelity in LLM-Compressed Financial Analysis (2026)

- Hoyoung Lee, Suhwan Park, **Seunghan Lee**, Jun Seo, Jaehoon Lee, Sungdong Yoo, Minjae Kim, CheolWon Na, Zhangyang Wang, Zach Golkhou, Minkyu Kim, Sotirios Sabanis, Alejandro Lopez-Lira, Dhagash Mehta, Soonyoung Lee, Chanyeol Choi, Wonbin Ahn, Yongjae Lee
- In *EMNLP, 2026.* [arxiv:2606.29251]

### 17. AdaTKG: Adaptive Memory for Temporal Knowledge Graph Reasoning (2026)

- **Seunghan Lee**, Jun Seo, Jaehoon Lee, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, SoonYoung Lee, Wonbin Ahn
- In *KDDW (GMLLM), 2026.* Oral [arxiv:2605.07121]

### 16. FinSTaR: Towards Financial Reasoning with Time Series Reasoning Models (2026)

- **Seunghan Lee**, Jun Seo, Jaehoon Lee, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, SoonYoung Lee, Wonbin Ahn
- In *EMNLP, 2026.* [arxiv:2605.03460]
- *Preliminary version: KDDW, 2026.* Oral

### 15. Not All Retrievals are Useful: Cross-Attention for Input-Aware RAG in Time Series Forecasting (2026)

- **Seunghan Lee**, Jaehoon Lee, Jun Seo, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, SoonYoung Lee, Wonbin Ahn
- In *KDDW (MILETS), 2026.* Oral [arxiv:2603.14709]

### 14. Rethinking Multimodal Fusion for Time Series: Auxiliary Modalities Need Constrained Fusion (2026)

- **Seunghan Lee**, Jun Seo, Jaehoon Lee, Sungdong Yoo, Minjae Kim, Tae Yoon Lim, Dongwan Kang, Hwanil Choi, SoonYoung Lee, Wonbin Ahn
- In *KDDW (MILETS), 2026.* Oral [arxiv:2603.22372]

### 13. FinTexTS: Financial Text-Paired Time-Series Dataset via Semantic-Based and Multi-Level Pairing (2026)

- Jaehoon Lee, Suhwan Park, Tae Yoon Lim, **Seunghan Lee**, Jun Seo, Dongwan Kang, Hwanil Choi, Minjae Kim, Sungdong Yoo, SoonYoung Lee, Yongjae Lee, Wonbin Ahn
- In *KDD, 2026.* [arxiv:2603.02702]

### 12. Mitigating Label Shift in Tabular In-Context Learning via Test-Time Posterior Adjustment (2026)

- **Seunghan Lee**
- In *ICML, 2026.* [arxiv:2605.04363]

### 11. Dataset-Driven Channel Masks in Transformers for Multivariate Time Series (2026)

- **Seunghan Lee**, Taeyoung Park, Kibok Lee
- In *ICASSP, 2026.* [arxiv:2410.23222]

### 10. Soft Contrastive Learning for Irregular Time Series (2025)

- Junghoon Lim, **Seunghan Lee**, Taeyoung Park
- In *ICMLW (Foundation Models for Structured Data), 2025.*

### 9. Channel Normalization for Time Series Channel Identification (2025)

- **Seunghan Lee**, Taeyoung Park[†], Kibok Lee[†]
- In *ICML, 2025.* [arxiv:2506.00432]

### 8. Sequential Order-Robust Mamba for Time Series Forecasting (2024)
- **Seunghan Lee**[*], Juri Hong[*], Kibok Lee[†], Taeyoung Park[†]
- In *NeurIPSW (Time Series in the Age of Large Models) , 2024.* [arxiv:2410.23356]

### 7. Partial Channel Dependence with Channel Masks for Time Series Foundation Models (2024)
- **Seunghan Lee**, Taeyoung Park[†], Kibok Lee[†]
- In *NeurIPSW (Time Series in the Age of Large Models), 2024.* Oral Presentation (Top 5) [arxiv:2410.23222]

### 6. Adaptive Noise Schedule for Time Series Diffusion Models (2024)
- **Seunghan Lee**, Kibok Lee[†], Taeyoung Park[†]
- In *NeurIPS, 2024.* [arxiv:2410.14488]
- Outstanding Paper Award from the **Journal of Korean Artificial Intelligence Association**

### 5. Learning to Embed Time Series Patches Independently (2024)
- **Seunghan Lee**, Taeyoung Park, Kibok Lee
- In *ICLR, 2024.* [arxiv:2312.16427]
- *Preliminary version: NeurIPSW (Self-Supervised Learning: Theory and Practice), 2023.* Oral Presentation (Top 4)

### 4. Soft Contrastive Learning for Time Series (2024)
- **Seunghan Lee**, Taeyoung Park, Kibok Lee
- In *ICLR, 2024.* Spotlight (366/7262=5%) [arxiv:2312.16424]
- *Preliminary version: NeurIPSW (Self-Supervised Learning: Theory and Practice), 2023.*

### 3. Hierarchical Multi-Task Learning with Self-Supervised Auxiliary Task (2024)
- **Seunghan Lee**, Taeyoung Park
- In *The Korean Journal of Applied Statistics 37(5): 631-641, 2024.*

### 2. MAD-GL2: Multimodal Adaptive Dynamic Graph Learning with Global and Local Features for Multivariate Time Series Forecasting (2025+)
- **Seunghan Lee**[*], Kibok Lee[*], Taeyoung Park
- Under review

### 1. Improving Gibbs Sampler (2022)
- Taeyoung Park, **Seunghan Lee**
- In Wiley Interdisciplinary Reviews: Computational Statistics 14(2): e1546, 2022.
- Number of Citations (Google Scholar, 2025.02.16): 11

## 4. Careers (Details)

**1. LG AI Research**

- **Period**: December 2025 –
- **Role**: AI Scientist
- **Division**: Data Intelligence Lab > Business Intelligence AI (BIAI)
- **Tasks**:
  - ① EXAONE Finance: TS foundation model for financial assets
  - ② EXAONE Demand: TS foundation model for demand forecasting
  - ③ Financial TS forecasting (Equity, Macro, Fundamental, etc.)
  - ④ Multimodal TS forecasting (TS + Text)
  - ⑤ Supply Chain Knowledge Graph
  - ⑥ TS reasoning model
  - ⑦ Explainable TS forecasting
- **Key Algorithms**: TS Foundation Models, Multimodal Learning, LLM Agents, Knowledge Graph
- **Programming Language**: Python

**2. KRAFTON**

- **Period**: May 2025 – December 2025
- **Role**: Deep Learning Engineer/Researcher
- **Division**: Deep Learning Division > AI Research > AI Agent Dept. > User Modeling AI Team
- **Tasks**:
  - ① PUBG anti-cheat modeling with tabular and time-series DL
  - ② PUBG Global Championship (PGC) win-rate prediction with tabular DL and survival analysis
- **Key Algorithms**: Tabular DL, Time Series DL, Survival Analysis
- **Programming Language**: Python

**3. Naver Cloud (internship)**

- **Period**: March 2025 – May 2025
- **Division**: HyperClova X
- **Tasks**: Multi-modal Pretraining.
  - ① Model training tasks for adding vision capabilities to a multimodal backbone
  - ② Exploration of the impact of various multimodal backbones on performance
  - ③ Efficient pretraining recipe exploration
- **Key Algorithms**: LLaVA, Qwen, DeepSeek
- **Programming Language**: Python

**4. SK Telecom (internship)**

- **Period**: July 2019 – August 2019
- **Division**: ICT Infra Center > Data Science > Data Analytics (AI Home)
- **Tasks**: Data analysis and service planning for SKT's air quality info service.
  - ① Analyzed air quality data and addressed measurement errors.
  - ② Designed strategies based on data analysis of parents in their 30s.
  - ③ Planned air quality reports, including data visualization.
- **Key Algorithms**: Clustering, DNN
- **Programming Language**: Python

**5. Yonsei Univ. Department of Computational Science & Engineering**

- **Period**: December 2019 – February 2020
- **Division**: Data Science & Deep Learning > Network Embedding
- **Tasks**: Implemented research papers and conducted related seminars.
- **Key Algorithms**: Graph Neural Network
- **Programming Language**: Python

## 5. Industry-Academia Collaboration

**1. Amore Pacific**

- **Period**: February 2020 – December 2022
- **[2020 Project]** User and purchase behavior analysis for Amore Pacific.
  - ① Analyzed purchasing behavior and conducted customer segmentation.
  - ② Predicted sales of products.
  - ③ Analyzed web log data to understand online user behavior patterns.
  - **Key Algorithms**: Random Forest, Logistic Regression, Hierarchical Clustering
- **[2021–2022 Project]** Development of Context Brand Score (CBS).
  - Developed a brand index that considers consumer sentiment on cosmetic product attributes using text data (social buzz, reviews).
  - **Key Algorithms**: Hierarchical BERT, Aspect-Based Sentiment Analysis.
- **Programming Language**: Python, R

**2. Hyundai Mobis**

- **Period**: June 2022 – December 2022
- **Tasks**:
  - ① Short- and long-term demand forecasting for low-circulation and eco-friendly parts of vehicles (Zero-inflated TS).
  - ② Utilizing textual data and tabular data for TS forecasting.
- **Key Algorithms**: GNN, GAN, Domain Adaptation, SSL with Tabular Data
- **Programming Language**: Python

## 6. Academic Service

- **Reviewer**: ICLR (2025, 2026, 2027), ICML (2024, 2025, 2026), NeurIPS (2024, 2025, 2026), AAAI (2025, 2026, 2027), EMNLP (2026), TMLR (2026)

## 7. Invited Talks

- 2024.08. Korean Artificial Intelligence Association (2024 Summer Conference)

## 8. Intellectual Property

- 2023.10: Review evaluation method using review quality metrics
- 2024.12: Universal time series analysis method and system for irregular multivariate time series

## 9. Programming Languages

- **Proficient**: Python (Pytorch, Tensorflow, Numpy, Sklearn)
- **Intermediate**: SQL, R
- **Basic**: Docker, Kubernetes

## 10. Instructor, Teaching Assistant (TA)

**1. [Instructor] Yonsei University Atmospheric Science Department – Data Analysis**

- **Period**: November 2020 – February 2021
- **Tasks**: Provided statistical and clustering analysis training for graduate students and assisted with data analysis.

**2. [Instructor] KB Bank – Computer Vision / GAN**

- **Period**: July 2021
- **Tasks**: Provided a lecture on computer vision and generative models

**3. [Instructor] Hanwha Ocean – Data Analysis & Statistics**

- **Period**: July 2023
- **Tasks**: Provided a lecture on below contents regarding statistical analysis

**4. [TA] Python Programming & Web Crawling**

- **Period**: March, September 2021,2022,2023,2024
- **Tasks**: Teaching assistant for Python programming and web crawling

**5. [TA] SK Hynix – Bayesian Statistics & ML Course**

- **Period**: April 2021 – June 2021
- **Tasks**: Conducted coding exercises for Bayesian Statistics and ML.

## 11. Academic Society & Club

**1. ISSU (Yonsei University IT Management Strategy Society)**

- **Period**: March 2018 – December 2018
- **Regular Sessions**: Conducted industry-academia collaboration projects with IT companies, including Naver Papago, Naver VIBE, Send Anywhere, and Amanda.
- **Study Sessions**:
  - 1st Semester: Contributed an article on AI in the Arts to IT Chosun (Link).
  - 2nd Semester: Served as a Python mentor.

**2. BITAMIN (Big Data Analytics Club)**

- **Period**: August 2018 – March 2019
- **Regular Sessions**: Focused on machine learning and text mining.
- **Study Sessions**: Conducted three machine learning projects.

**3. Data Science Lab (Yonsei University Data Science Society)**

- **Period**: February 2019 – June 2020
- **The 1st President**: Organized regular sessions and study groups on ML and DL.
- **Regular Sessions**:
  - ① Studied the entire data science workflow, from data collection and preprocessing to modeling.
  - ② Conducted training sessions for society members on:
    - Data Preprocessing, Association Analysis, SVM, Boosting, CNN
- **Study Sessions**: Group projects and data science competitions.

| Name | Details | Period | Etc. |
|---|---|---|---|
| ISSU | Yonsei Univ. IT Management Strategy Society | 2018.03 - 2018.12 | Industry-academia collaboration projects. Python mentor. |
| Bitamin | Big Data Analytics Club | 2018.07 - 2019.06 | |
| Data Science Lab | Yonsei Univ. Data Science Society | 2019.01 - 2020.06 | The 1st President |

**4. OWOP (One Week One Paper)**

- **Period**: January 2021 – June 2021
- **Tasks**: Conducted a research paper study group with five graduate students from the Department of Statistics and Data Science at Yonsei University.
- **Topics**: AI and statistics-related subjects, including CV, NLP, BNN, Recommender Systems, Deep Generative Models (VAE, Normalizing Flow).

**5. Deep Learning Paper Reading Study**

- **Period**: May 2021 – September 2021
- **Tasks**: Conducted a research paper study group with five graduate students from the Department of Statistics and Data Science at Yonsei University.
- **Topics**: Meta Learning, Continual Learning, Interpretable/Reliable Learning

## 12. Certifications & Awards

### Certifications

| Field | Certifications | Score | Date |
|---|---|---|---|
| CS/Data | ADSP (Associate Data Science Professional) | - | 2018.09 |
| | Computer Proficiency Level 1 | | 2019.03 |
| | SQLD (SQL Developer) | | 2020.06 |
| | Big Data Analyst | | 2021.07 |
| | AWS Cloud Practitioner | | 2021.08 |
| Linguistics | TOEIC | 965/990 | 2020.08 |
| Etc. | Korean History Proficiency Test | Level 1 | 2012.08 |
| | TESAT (Korean Economic Newspaper) | S-Level | 2013.05 |

### Awards

| Field | Awards | Rank | Date |
|---|---|---|---|
| Academic | Academic Excellence Awards | | 2018.12 / 2019.06 |
| | Summa Cum Laude Graduation | | 2020.08 |
| CS | Yonsei Data Science Competition | 2nd | 2019.11 |
| | Sony Pictures Audience Prediction | 4th | 2020.01 |
| | Yonsei Data Science Competition | 2nd | 2021.12 |
| | BK Winter Academic Conference | 4th | 2022.12 |
| | Yonsei Data Science Competition | 1st | 2023.02 |
| Economy | TESAT Team Excellence Award | 2nd | 2013.05 |

## 13. Projects

**1. Fine-tuning LLM**

- **Period**: February 2025 – March 2025
- **Tasks**:
  - (1) Fine-tuning LLaMA with Q-LoRA (Single-GPU)
  - (2) Fine-tuning LLaMA with Q-LoRA & FFT (Multi-GPU)
    - Axolotl, code-based
    - Distributed training: MP, FSDP, ZeRO
  - (3) Fine-tuning LLaMA for the insurance domain
    - One-cycle project: Data construction – SFT – DPO – Inference
    - Extract text from PDFs using OCR, generate additional questions with WizardLM (i.e., evolving), and build SFT & DPO datasets

**2. Hands-on practice with Ollama, RAG pipeline, Vector DB, and UI/UX tools**

- **Period**: February 2025 – March 2025
- **Tasks**:
  - (1) Ollama + RAG (PDF upload) + ChromaDB + Streamlit
  - (2) Ollama + RAG (Repository) + ChromaDB + Chainlit
  - (3) Ollama + RAG (Repository) + ChromaDB + Gradio

## 14. Competitions

**1. [Dacon] 2019 KBO Baseball Player Performance Prediction**

- **Period**: January 2019 – February 2019
- **Tasks**: Predicting KBO baseball player performance for the first half of 2019.
- **Key Algorithms**: MA, ARIMA, Random Forest

**2. [Yonsei Data Science Competition] Embrain**

- **Period**: March 2019 – May 2019
- **Tasks**: Developing a marketing strategy for highly meticulous consumers.
- **Key Algorithms**: Random Forest, NN
- **Result**: 4th

**3. [Yonsei Data Science Competition] PACKUS**

- **Period**: September 2019 – November 2019
- **Tasks**:
  - ① Customer segmentation and personalized strategy development.
  - ② Future sales prediction for key products.
  - ③ Created a dashboard for CRM using R Shiny.
- **Key Algorithms**: RNN, XGBoost, LightGBM, SOM, Association Analysis
- **Result**: 2nd

**4. [Sony Pictures] Movie Audience Prediction**

- **Period**: November 2019 – January 2020
- **Tasks**: Predicting cumulative audience numbers for the first 14 days after a movie's release.
- **Key Algorithms**: Random Forest, XGBoost, Stacking, NN
- **Result**: 4th

**5. [Kakao] Melon Playlist Continuation**

- **Period**: May 2019 – July 2019
- **Tasks**: Predicting missing songs and tags in a given playlist.
- **Key Algorithms**: Collaborative Filtering, Spectrogram Analysis

**6. [IGA Works] CTR Prediction**

- **Period**: December 2019 – February 2020
- **Tasks**: Predicting click-through probability when a user is exposed to an ad.
- **Key Algorithm**: Deep CTR

**7. [Dacon] AI Competition for Psychological Tendency Prediction**

- **Period**: September 2020 – November 2020
- **Tasks**: Developing an algorithm for psychological test analysis and voter prediction.
- **Key Algorithms**: AutoML, Deep CTR

**8. [BigCon Test] NS Home Shopping Schedule Optimization**

- **Period**: July 2020 – October 2020
- **Tasks**: Optimizing home shopping schedules to maximize sales.
- **Key Algorithms**: CatBoost, LightGBM, Bayesian Optimization

**9. [Hyundai Industries] Big Data/AI Competition**

- **Period**: January 2021 – February 2021
- **Tasks**: Predicting manufacturing process task durations and optimizing task allocation.
- **Key Algorithms**: Bayesian Neural Network, LightGBM
- **Result**: Finalist (18th place out of 284 teams)

**10. [Yonsei Data Science Competition] Hyodol**

- **Period**: September 2021 – December 2021
- **Tasks**: Customer clustering and predicting optimal engagement times for personalized care.
- **Key Algorithms**: Multi-task Learning, Self-Organizing Map
- **Result**: 2nd

**11. [Conference] 2022 Winter BK Academic Conference**

- **Period**: December 2022
- **Tasks**: Multivariate time series forecasting using Spatio-Temporal GNN.
- **Key Algorithms**: Spatio-Temporal GNN
- **Result**: 4th

**12. [Yonsei Data Science Competition] KCB**

- **Period**: December 2022 – February 2023
- **Tasks**: Multi-task learning with self-supervised learning.
    - ① Regression: Credit score prediction
    - ② Classification: Credit rating prediction
    - ③ Time Series Forecasting: Future revenue prediction
    - ④ Clustering: Customer segmentation
- **Key Algorithms**: SSL with Tabular Data
- **Result**: 1st